Gavin's Lectures on Physics - Optics



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Contents

1 Geometric Optics **3**
- 1.1 Fermat's Principle . . . 3
- 1.2 Snell's Law . . . 3

2 Interference **4**
- 2.1 Types of Interference . . . 4
- 2.2 Intensity of Interference . . . 4
- 2.3 Superposition of Waves . . . 5
  - 2.3.1 Incoherent case . . . 5
  - 2.3.2 Coherent case . . . 5
- 2.4 Interference between Two Coherent Point Sources . . . 5
- 2.5 Real Field and Multi-Source Interference . . . 6
- 2.6 Interference between Multiple Coherent Point Sources (Point-Source Model) 6

3 Diffraction **8**
- 3.1 Green's Formula . . . 8
- 3.2 Kirchhoff Formula . . . 8
- 3.3 Green Function of the Helmholtz Equation . . . 8
- 3.4 Huygens–Fresnel Diffraction . . . 9
- 3.5 Fresnel Number . . . 11
- 3.6 Fraunhofer Diffraction . . . 11
  - 3.6.1 Geometric setup and Fraunhofer approximation . . . 11
  - 3.6.2 Fraunhofer Diffraction from a Rectangular Aperture . . . 12
- 3.7 Fresnel Diffraction . . . 12
  - 3.7.1 Geometric setup and Fresnel approximation . . . 12
  - 3.7.2 Amplitude on the axial point . . . 13
  - 3.7.3 Cornu spiral . . . 13

4 Diffraction in Optical Instruments **15**
- 4.1 Young's Double Slit (Equal Widths) . . . 15
- 4.2 Double Slit Interference with Unequal Slit Widths (General Case) . . . 15
- 4.3 Diffraction in Optical Instruments . . . 16
  - 4.3.1 Simple Imaging Lens . . . 16
  - 4.3.2 Telescope . . . 16
  - 4.3.3 Microscope . . . 16
- 4.4 Microscope Resolution . . . 17
- 4.5 Diffraction Grating . . . 17

5 Interferometry **19**
- 5.1 Interference Fringes . . . 19
  - 5.1.1 Equal inclination fringes . . . 19
  - 5.1.2 Equal thickness fringes . . . 19
- 5.2 The Michelson Interferometer . . . 19
- 5.3 The Fabry–Perot Etalon and Spectrometer . . . 21

6 Fundamentals of Lasers **23**
6.1 Interaction of Atoms . . . 23
6.2 Conditions for Laser Gain . . . 23
6.3 Laser Longitudinal Modes . . . 24
6.3.1 Resonance Condition . . . 24
6.3.2 Frequency Separation of Consecutive Longitudinal Modes . . . 24
6.3.3 Wavelength Linewidth . . . 24
6.3.4 Number of Lasing Longitudinal Modes . . . 24

7 Coherence Theory **25**
7.1 Coherence . . . 25
7.1.1 Temporal Coherence . . . 25
7.1.2 Spatial (Transverse) Coherence . . . 25
7.2 Extended Source Spatial Coherence . . . 26
7.3 Van Cittert–Zernike Theorem . . . 26
7.3.1 Astronomical Imaging Interference . . . 27

8 Polarization of Light **28**
8.1 General Formulation . . . 28
8.2 Polarization States . . . 28
8.3 Jones Vectors . . . 28
8.4 Jones Matrices . . . 28
8.5 Stokes Parameters . . . 29
8.6 Malus's Law, Brewster Angle, Faraday Effect . . . 29
8.6.1 Malus's Law . . . 29
8.6.2 Brewster's Angle . . . 29
8.6.3 Faraday Effect . . . 29
8.7 Polarization Optical Components and Birefringence . . . 29
8.7.1 Polarization Optical Components . . . 29
8.7.2 Birefringence . . . 30

1 Geometric Optics

1.1 Fermat's Principle

Fermat's principle states that the actual path taken by light between two points is the one for which the optical path length is stationary:

$$0 = \delta \int_S \mathrm{d}t = \frac{1}{c}\,\delta \int_S n\,\mathrm{d}s = \frac{1}{c}\,\delta L, \tag{1}$$

where the optical path length is defined as

$$L = \int_S n\,\mathrm{d}s. \tag{2}$$

1.2 Snell's Law

Consider the geometry with distances

$$l_1 = \frac{1}{\cos\theta_1}, \qquad l_2 = \frac{1}{\cos\theta_2}, \tag{1}$$

and

$$L = \tan\theta_1 + \tan\theta_2. \tag{2}$$

The travel time is

$$t = \frac{1}{c}\left(l_1 n_1 + l_2 n_2\right). \tag{3}$$

From (2) we have

$$\delta\theta_2 = -\frac{\cos^2\theta_2}{\cos^2\theta_1}\,\delta\theta_1. \tag{3}$$

Applying Fermat's principle:

$$\begin{aligned} 0 = \delta \int_S \mathrm{d}t &= \frac{1}{c}\left(n_1\delta l_1 + n_2\delta l_2\right) \\ &= \frac{1}{c}\left(n_1\delta\frac{1}{\cos\theta_1} + n_2\delta\frac{1}{\cos\theta_2}\right) \\ &= \frac{1}{c}\left(n_1\frac{\sin\theta_1}{\cos^2\theta_1}\delta\theta_1 + n_2\frac{\sin\theta_2}{\cos^2\theta_2}\delta\theta_2\right) \\ &= \frac{1}{c}\left(n_1\sin\theta_1 - n_2\sin\theta_2\right)\cdot\frac{1}{\cos^2\theta_1}\cdot\delta\theta_1. \end{aligned} \tag{4}$$

Thus we obtain Snell's law:

$$n_1\sin\theta_1 = n_2\sin\theta_2. \tag{5}$$

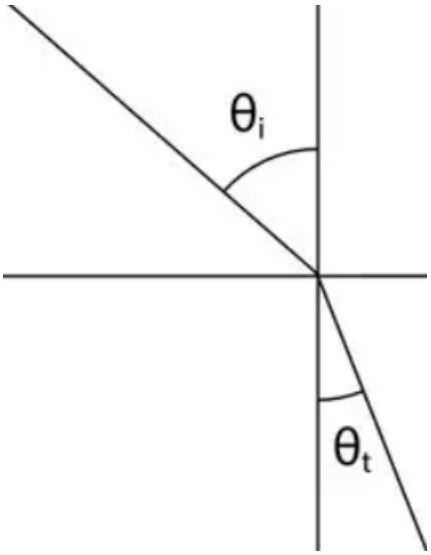


Figure 1: Reflection and refraction of a light ray at a dielectric interface, illustrating Snell's law and the law of reflection.

2 Interference

2.1 Types of Interference

Wave-front division : Splits one wavefront into coherent sub-sources that interfere in space.

Examples: Young' s double slit, Fresnel biprism, Lloyd' s mirror.

Amplitude division : Splits beam amplitude via film reflections; interference comes from optical path difference.

Examples: Equal-inclination interference, wedge film interference, Newton' s rings.

2.2 Intensity of Interference

The real wave is expressed as

$$u = u_0 \cos\left(\omega t - \mathbf{k} \cdot \mathbf{r}\right). \tag{6}$$

Using complex representation:

$$u = u_0 e^{i(\omega t - \mathbf{k}\cdot\mathbf{r})}, \qquad u_0 = |u_0| e^{i\phi}, \tag{7}$$

and the conjugate

$$u^* = u_0^* e^{-i(\omega t - \mathbf{k}\cdot\mathbf{r})}, \qquad u_0^* = |u_0| e^{-i\phi}. \tag{8}$$

The intensity (irradiance) is proportional to

$$I \propto \langle u u^* \rangle. \tag{9}$$

2.3 Superposition of Waves

The superposition at point P is

$$u_P = u_A + u_B, \tag{10}$$

with

$$u_A = u_{A0} e^{i(\omega t - k_A r_A)}, \qquad u_B = u_{B0} e^{i(\omega t - k_B r_B)}. \tag{11}$$

The intensity expansion gives

$$I_P \propto \langle u_P u_P^* \rangle = \langle u_A u_A^* \rangle + \langle u_B u_B^* \rangle + \langle u_A u_B^* + u_A^* u_B \rangle. \tag{12}$$

2.3.1 Incoherent case

If $k_A \neq k_B$, $\omega_A \neq \omega_B$ (unrelated sources), then

$$\langle u_A u_B^* \rangle = \langle u_A \rangle \langle u_B^* \rangle = 0, \tag{13}$$

hence

$$I_P = I_A + I_B. \tag{14}$$

2.3.2 Coherent case

If $k_A = k_B$, $\omega_A = \omega_B$, then

$$I_P = I_A + I_B + 2\sqrt{I_A I_B} \cos\left(\phi_A - \phi_B\right). \tag{15}$$

2.4 Interference between Two Coherent Point Sources

The field at point P is

$$u_P = \frac{u_{0A}}{r_A} e^{i(\omega t - k r_A)} + \frac{u_{0B}}{r_B} e^{i(\omega t - k r_B)}. \tag{16}$$

In the far-field regime ($r_A, r_B \gg \lambda$), we use the approximations

$$r_A \approx r - \frac{x \sin\theta}{2}, \qquad r_B \approx r + \frac{x \sin\theta}{2}. \tag{17}$$

Assuming equal amplitudes $u_{0A} = u_{0B} = u_0$, we get

$$\begin{aligned} u_P &= \frac{u_0}{r} e^{i(\omega t - kr)} \left[e^{ik\frac{x\sin\theta}{2}} + e^{-ik\frac{x\sin\theta}{2}} \right] \\ &= 2\frac{u_0}{r} e^{i(\omega t - kr)} \cos\left(\frac{\delta}{2}\right), \end{aligned} \tag{18}$$

with phase difference

$$\delta = kx \sin\theta. \tag{19}$$

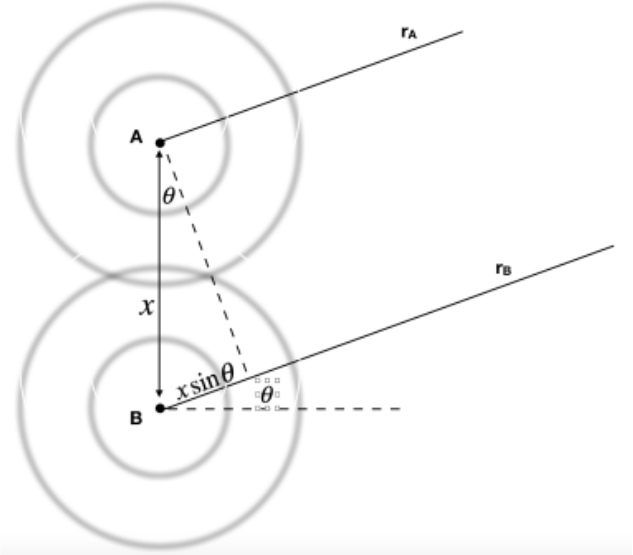


Figure 2: Geometry of two-point-source interference: two coherent sources S_1 and S_2 emit spherical waves that interfere at point P on a distant screen.

2.5 Real Field and Multi-Source Interference

The real physical field is

$$u_{P,\mathrm{Re}} = \mathrm{Re}(u_P) = 2\frac{u_0}{r}\cos(\omega t - kr)\cos\left(\frac{\delta}{2}\right). \tag{20}$$

Time-averaged intensity:

$$\begin{aligned} I \propto \langle u_{P,\mathrm{Re}}^2 \rangle &= 4\frac{u_0^2}{r^2}\cos^2\left(\frac{\delta}{2}\right)\langle \cos^2(\omega t - kr)\rangle \\ &= 2\frac{u_0^2}{r^2}\cos^2\frac{\delta}{2} = \frac{u_0^2}{r^2}(1+\cos\delta). \end{aligned} \tag{21}$$

Thus

$$I = I_0\cos^2\left(\frac{kx\sin\theta}{2}\right). \tag{22}$$

For incoherent sources, $\langle\cos\delta\rangle = 0$, and $I \propto \frac{u_0^2}{r^2}$.

2.6 Interference between Multiple Coherent Point Sources (Point-Source Model)

For N equally spaced point sources (no slit width),

$$u \propto \sum_{m=0}^{N-1} e^{im\delta}. \tag{23}$$

Summing the geometric series:

$$\begin{aligned} u &= \frac{e^{iN\delta} - 1}{e^{i\delta} - 1} \\ &= e^{i(N-1)\delta/2} \, \frac{\sin\left(\frac{N\delta}{2}\right)}{\sin\left(\frac{\delta}{2}\right)}. \end{aligned} \tag{24}$$

The intensity is

$$I(\delta) \propto \langle uu^* \rangle = \frac{I_0}{N^2} \, \frac{\sin^2\left(\frac{N\delta}{2}\right)}{\sin^2\left(\frac{\delta}{2}\right)}. \tag{25}$$

Maxima occur when $\delta = 2n\pi$, yielding $I_{\max} = I_0$. This result will be used later in the diffraction grating analysis (Section 4.6), where finite slit widths are taken into account.

3 Diffraction

3.1 Green's Formula

The vector calculus identity is

$$\nabla \cdot (\psi \nabla \varphi) = \nabla \psi \cdot \nabla \varphi + \psi \nabla^2 \varphi. \tag{26}$$

Subtracting the symmetric form gives Green's second identity:

$$\int_V \left(\psi \nabla^2 \varphi - \varphi \nabla^2 \psi\right) \mathrm{d}V = \oint_S \left(\psi \frac{\partial \varphi}{\partial n} - \varphi \frac{\partial \psi}{\partial n}\right) \mathrm{d}s = \oint_S \left(\psi \nabla \varphi - \varphi \nabla \psi\right) \cdot \mathrm{d}\mathbf{S}. \tag{27}$$

3.2 Kirchhoff Formula

With $k = \frac{\omega}{c}$, the wave equations for the electromagnetic field are

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right) \mathbf{E} = 0, \qquad \left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right) \mathbf{B} = 0. \tag{28}$$

Monochromatic plane wave solutions:

$$\mathbf{E} = E_0 \cos\left(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta\right) \hat{n}, \tag{29}$$

$$\mathbf{B} = \frac{1}{c} E_0 \cos\left(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta\right) \left(\hat{k} \times \hat{n}\right). \tag{30}$$

Reduction to the Helmholtz equation:

$$\left(\nabla^2 + k^2\right) \psi = 0, \tag{31}$$

where ψ represents any Cartesian component of $\mathbf{E}$ or $\mathbf{B}$.

3.3 Green Function of the Helmholtz Equation

Let $G(\mathbf{R})$ be the Green function satisfying

$$\left(\nabla^2 + k^2\right) G(\mathbf{R}) = -\delta(\mathbf{r}_s - \mathbf{r}_0) = -\delta(\mathbf{R}), \tag{32}$$

with $\mathbf{R} = \mathbf{r}_s - \mathbf{r}_0$. For spherical symmetry, $G = G(R)$, and the radial ODE is

$$\frac{1}{R^2}\frac{\mathrm{d}}{\mathrm{d}R}\left(R^2 \frac{\mathrm{d}G}{\mathrm{d}R}\right) + k^2 G = 0. \tag{33}$$

The outgoing solution is

$$G(R) = \frac{e^{ikR}}{4\pi R}. \tag{34}$$

Applying Green's second identity to ψ and G:

$$G\left(\nabla^2 + k^2\right) \psi = 0, \tag{35}$$

$$\psi\left(\nabla^2 + k^2\right) G = -\psi(\mathbf{r}_s)\delta(\mathbf{r}_s - \mathbf{r}_0). \tag{36}$$

Thus

$$\begin{aligned}\psi(\mathbf{r}_0) &= \int \psi(\mathbf{r}_s)\delta(\mathbf{r}_s - \mathbf{r}_0)\,\mathrm{d}V \\ &= \int \left(G\nabla^2\psi - \psi\nabla^2 G\right)\mathrm{d}V \\ &= \oint_S \left(G\frac{\partial\psi}{\partial n} - \psi\frac{\partial G}{\partial n}\right)\mathrm{d}s. \end{aligned} \tag{37}$$

This yields the full Kirchhoff integral:

$$\psi(\mathbf{r}_0) = \oint_S \left[G(\mathbf{r},\mathbf{r}_0)\nabla\psi(\mathbf{r}_s) - \psi(\mathbf{r}_s)\nabla G(\mathbf{r},\mathbf{r}_0)\right]\cdot\mathrm{d}\mathbf{S} = \frac{1}{4\pi}\oint_S \left[\frac{e^{ikr}}{r}\nabla\psi - \psi\nabla\frac{e^{ikr}}{r}\right]\cdot\mathrm{d}\mathbf{S}. \tag{38}$$

The gradient of the Green function is

$$\nabla G = \nabla\frac{e^{ikr}}{4\pi r} = \frac{\hat{r}}{4\pi}\frac{\partial}{\partial r}\left(\frac{e^{ikr}}{r}\right) = \frac{e^{ikr}}{4\pi r}\left(ik - \frac{1}{r}\right)\hat{r}. \tag{39}$$

In the far-field limit ($kr \gg 1$):

$$\nabla G \approx \frac{e^{ikr}}{4\pi r}\,ik\,\hat{r} = ikG\,\hat{r}, \qquad \mathrm{d}\mathbf{S} = \hat{n}\,\mathrm{d}s. \tag{40}$$

3.4 Huygens–Fresnel Diffraction

For a spherical source,

$$\psi(\mathbf{r}) = \frac{Ae^{ik|\mathbf{r}-\mathbf{r}_{\text{source}}|}}{|\mathbf{r}-\mathbf{r}_{\text{source}}|} = \frac{Ae^{ikr_s}}{r_s}. \tag{41}$$

In the far field, $\nabla\psi \approx ik\psi\,\hat{r}_s$.

Huygens–Fresnel Diffraction Formula Starting from the Kirchhoff integral:

$$\begin{aligned}\psi(\mathbf{r}_0) &= \oint_S (G\nabla\psi - \psi\nabla G)\cdot\hat{n}\,\mathrm{d}s \\ &\approx \oint_S \left[ik\left(G\psi\,\hat{r}_s + \psi G\,\hat{r}\right)\cdot\hat{n}\right]\mathrm{d}s \\ &= \frac{ik}{4\pi}\oint_S \psi(\mathbf{r})\frac{e^{ikr}}{r}\left(\cos\theta_s + \cos\theta\right)\mathrm{d}s \\ &= \frac{1}{i\lambda}\oint_S \psi(\mathbf{r})\frac{e^{ikr}}{r}\,\frac{\cos\theta + \cos\theta_s}{2}\,\mathrm{d}s. \end{aligned} \tag{42}$$

For normal incidence ($\theta_s = 0$, $\hat{r}_s = \hat{n}$, $\hat{r}\cdot\hat{n} = \cos\theta$):

$$\psi(\mathbf{r}_0) = \frac{1}{i\lambda}\oint_S \psi(\mathbf{r})\frac{e^{ikr}}{r}\,\frac{1+\cos\theta}{2}\,\mathrm{d}s. \tag{43}$$

In compact notation:

$$\psi(a) = \frac{1}{i\lambda}\oint_S \psi(\rho)\frac{e^{ikr}}{r}\left(\frac{1+\cos\theta}{2}\right)\mathrm{d}s. \tag{44}$$

For a spherical source with boundary conditions

$$\psi_P = \psi(a) = \psi_0 \frac{e^{ika}}{a}, \qquad \psi(\rho) = \psi_0 \frac{e^{ik\rho}}{\rho}, \qquad \psi_A = \psi_0, \tag{45}$$

and in the paraxial limit ($a \gg \rho$, $\alpha \approx \theta$), we have

$$\frac{1}{r} \approx \frac{1}{a}, \qquad \cos\theta \approx \cos\alpha = \frac{x}{\rho}, \tag{46}$$

with the geometrical substitutions

$$x = \rho\cos\alpha, \qquad r = a - x, \qquad \mathrm{d}s = 2\pi\rho\sin\alpha \cdot \rho\,\mathrm{d}\alpha = -2\pi\rho\,\mathrm{d}x. \tag{47}$$

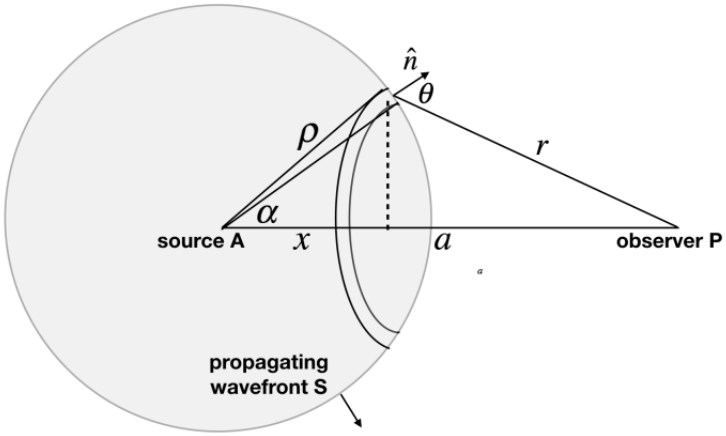


Figure 3: Huygens–Fresnel diffraction principle: each point on a wavefront acts as a secondary spherical wavelet source; the envelope of these wavelets forms the new wavefront.

Proof of the Huygens–Fresnel Integral

$$\begin{aligned}
\psi(a) &= \frac{1}{i\lambda} \oint \psi(\rho) \frac{e^{ikr}}{r} \frac{1+\cos\theta}{2}\,\mathrm{d}s \\
&\approx \frac{1}{i\lambda} \int_{-\rho}^{\rho} \frac{\psi_0 e^{ik\rho}}{\rho} \cdot \frac{e^{ik(a-x)}}{a} \cdot \frac{1+x/\rho}{2} \cdot 2\pi\rho\,\mathrm{d}x \\
&= \frac{\pi}{i\lambda} \cdot \psi_0 \frac{e^{ik(\rho+a)}}{\rho\, a} \int_{-\rho}^{\rho} e^{-ikx}(x+\rho)\,\mathrm{d}x \\
&= \frac{\pi}{i\lambda} \psi_0 \frac{e^{ik(\rho+a)}}{\rho\, a} \cdot \left(-\frac{1}{ik}\right) \cdot 2\rho \left(e^{-ik\rho} - \frac{\sin k\rho}{k\rho}\right).
\end{aligned} \tag{48}$$

In the limit $k\rho \gg 1$:

$$\psi(a) \approx \frac{\pi}{i\lambda} \psi_0 \frac{e^{ik(\rho+a)}}{\rho\, a} \cdot \left(-\frac{2\rho}{ik}\right) e^{-ik\rho} = \psi_0 \frac{e^{ika}}{a}. \tag{49}$$

3.5 Fresnel Number

The nature of diffraction observed at a given distance is determined by the Fresnel number, which quantifies the relative importance of the quadratic phase curvature across the aperture. Using the binomial expansion for small $x \ll r_0$:

$$\sqrt{1+x} \approx 1 + \frac{1}{2}x - \frac{1}{8}x^2. \tag{50}$$

The path length from a point in the aperture to the observation point is expanded as

$$r = \left(r_0^2 + x^2 - 2ux\right)^{1/2} \approx r_0 \left(1 - \frac{ux}{r_0^2} + \frac{x^2}{2r_0^2}\right). \tag{51}$$

Thus the phase factor becomes

$$\psi_P \propto e^{ikr} = e^{ikr_0} \cdot e^{-ikux/r_0} \cdot e^{ikx^2/(2r_0)}. \tag{52}$$

The Fresnel number is defined as

$$F = \frac{x^2}{\lambda r_0} = \frac{1}{2\pi}\frac{kx^2}{r_0}. \tag{53}$$

The distinction between the two diffraction regimes is based on the value of F:

- **Fraunhofer (far-field) diffraction** occurs when $F \ll 1$. In this limit, the quadratic phase term $e^{ikx^2/(2r_0)}$ is negligible across the aperture ($r_A \approx r_B$), and the phase variation reduces to a linear function of x: $\delta \propto x$. The diffraction pattern is simply the Fourier transform of the aperture function.
- **Fresnel (near-field) diffraction** occurs when $F \sim 1$ or $F \gtrsim 1$. The quadratic phase term cannot be neglected, and the propagation must be treated using the Fresnel approximation (convolution with a quadratic phase factor), leading to more complex patterns such as those described by the Cornu spiral.

3.6 Fraunhofer Diffraction

3.6.1 Geometric setup and Fraunhofer approximation

Let the aperture have dimensions $a \times b$ in the x–y plane, and let the observation point be at (u, v) with distance R from the origin. Define $r_0^2 = R^2 + u^2 + v^2$. Then

$$r^2 = R^2 + (u-x)^2 + (v-y)^2 = r_0^2 - 2ux - 2vy + x^2 + y^2. \tag{54}$$

Using the binomial approximation under the Fraunhofer condition ($F \ll 1$):

$$\begin{aligned} r &= r_0\sqrt{1 - 2\frac{ux+vy}{r_0^2} + \frac{x^2+y^2}{r_0^2}} \\ &\approx r_0\left(1 - \frac{ux+vy}{r_0^2}\right) = r_0 - \frac{ux+vy}{r_0}. \end{aligned} \tag{55}$$

3.6.2 Fraunhofer Diffraction from a Rectangular Aperture

The elementary field contribution is

$$\mathrm{d}u_P \propto e^{ikr}\,\mathrm{d}x\,\mathrm{d}y = e^{ikr_0}e^{-ik\frac{ux+vy}{r_0}}\,\mathrm{d}x\,\mathrm{d}y. \tag{56}$$

Integrating over the aperture:

$$\begin{aligned} u_P &\propto e^{ikr_0}\int_{-a/2}^{a/2}\mathrm{d}x\int_{-b/2}^{b/2}\mathrm{d}y\; e^{-ik(ux+vy)/r_0} \\ &\propto e^{ikr_0}\,ab\cdot\frac{\sin\left(\frac{ka}{2}\frac{u}{r_0}\right)}{\frac{kau}{2r_0}}\cdot\frac{\sin\left(\frac{kb}{2}\frac{v}{r_0}\right)}{\frac{kbv}{2r_0}} \\ &\propto e^{ikr_0}\,ab\cdot\mathrm{sinc}\left(\frac{ka}{2}\frac{u}{r_0}\right)\mathrm{sinc}\left(\frac{kb}{2}\frac{v}{r_0}\right), \end{aligned} \tag{57}$$

where $\mathrm{sinc}(x) = \frac{\sin x}{x}$. Thus the intensity is

$$I(u,v) \propto |u_P|^2 \propto a^2b^2\mathrm{sinc}^2\left(\frac{ka}{2}\frac{u}{r_0}\right)\mathrm{sinc}^2\left(\frac{kb}{2}\frac{v}{r_0}\right). \tag{58}$$

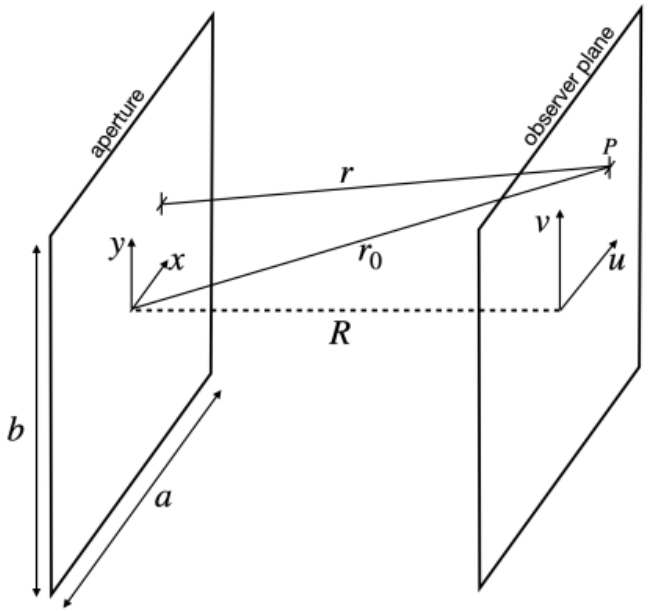


Figure 4: Fraunhofer diffraction through a rectangular aperture: the far-field intensity pattern is the product of two sinc functions, corresponding to diffraction along the two orthogonal axes.

3.7 Fresnel Diffraction

3.7.1 Geometric setup and Fresnel approximation

Consider a narrow slit of half-width a located in the y-direction, illuminated by a plane wave of wavelength λ ($k = 2\pi/\lambda$). The observation point P lies on a screen at distance z from the aperture, with transverse coordinate u. For the axial point Q, we have

$u = 0$. Under the standard Fresnel approximation ($z \gg a, |u|$), the distance r from an infinitesimal aperture element at y to the observation point is expanded as:

$$r = \sqrt{z^2 + (u-y)^2} \approx r_0 - \frac{uy}{r_0} + \frac{y^2}{2r_0}, \quad \text{where } r_0 = \sqrt{z^2+u^2}. \tag{59}$$

For the axial point Q ($u = 0$), this simplifies to $r_0 = z$, thus

$$r \approx z + \frac{y^2}{2z}. \tag{60}$$

3.7.2 Amplitude on the axial point

The diffracted field amplitude at the axial point Q is proportional to the integral of the phase over the slit. Let the slit have total width a and be centered on the y-axis, so the integration runs from $-a/2$ to $a/2$. Under the Fresnel approximation, the distance r from a point at y to Q is $r \approx z + y^2/(2z)$. Thus,

$$u_Q(z) \propto \int_{-a/2}^{a/2} e^{ikr}\, dy \propto e^{ikz} \int_{-a/2}^{a/2} e^{i\frac{ky^2}{2z}} dy = e^{ikz} \int_{-a/2}^{a/2} e^{i\frac{\pi y^2}{\lambda z}} dy. \tag{61}$$

To normalize, introduce the dimensionless variable

$$t = y\sqrt{\frac{2}{\lambda z}}, \qquad v = \frac{a}{2}\sqrt{\frac{2}{\lambda z}} \tag{62}$$

Substituting, the integral becomes

$$u_Q(z) \propto \int_{-v}^{v} e^{i\pi t^2/2}\, dt. \tag{63}$$

3.7.3 Cornu spiral

The Cornu spiral (also known as the clothoid) is the parametric curve in the complex plane defined by the Fresnel function:

$$Z(v) = C(v) + i\, S(v) = \int_0^v e^{i\pi t^2/2}\, dt, \tag{64}$$

where the coordinates are explicitly given by the Fresnel integrals:

$$C(v) = \int_0^v \cos\left(\frac{\pi t^2}{2}\right) dt, \qquad S(v) = \int_0^v \sin\left(\frac{\pi t^2}{2}\right) dt. \tag{65}$$

The parameter v represents the arc length along the curve (as will be shown below). The spiral winds towards two distinct focal points in the complex plane as $v \to \pm\infty$.

Property 1: Asymptotic limits The Cornu spiral converges to two finite points because the oscillatory integrand decays in the average sense. The limits of the complex Fresnel function are:

$$\lim_{v\to+\infty} Z(v) = \frac{1}{2} + i\frac{1}{2}, \qquad \lim_{v\to-\infty} Z(v) = -\frac{1}{2} - i\frac{1}{2}. \tag{66}$$

Geometrically, this means the spiral starts at the lower-left focal point $(-1/2, -i/2)$ when $v = -\infty$, spirals outwards through the origin $(0, 0)$ at $v = 0$, and finally converges to the upper-right focal point $(1/2, i/2)$ as $v \to +\infty$.

Property 2: Arc length The most elegant property of the Cornu spiral is that the parameter v exactly equals the arc length measured from the origin. Indeed, differentiating the complex function $Z(v)$ gives:

$$dZ = dC + i\,dS = \left(\cos\frac{\pi v^2}{2} + i\sin\frac{\pi v^2}{2}\right) dv = e^{i\pi v^2/2}\,dv. \tag{67}$$

The infinitesimal arc length ds along the curve is the modulus of this complex differential:

$$ds = \sqrt{dC^2 + dS^2} = \left|dC + i\,dS\right| = \left|e^{i\pi v^2/2}\right| |dv| = |dv|. \tag{68}$$

Therefore, the arc length from the origin $(0,0)$ to the point $(C(v), S(v))$ for $v > 0$ is:

$$s(v) = \int_0^v ds = \int_0^v |dv| = v. \tag{69}$$

Hence, v is not merely a mathematical parameter; it has a direct geometrical interpretation as the physical distance traveled along the spiral.

Property 3: Diffraction intensity and geometric chord Intensity $I \propto |u|^2$. And I_0 is the intensity for the fully open case. For an aperture spanning from v_1 to v_2, the intensity is:

$$I = \frac{1}{2} I_0 \left[\big(C(v_2) - C(v_1)\big)^2 + \big(S(v_2) - S(v_1)\big)^2\right]. \tag{70}$$

Therefore, the diffraction intensity is directly proportional to the square of the chord distance between the two corresponding points on the spiral. For the symmetric case where $v = v_2 = -v_1$, we have:

$$I = 2I_0 \left[C^2(v) + S^2(v)\right]. \tag{71}$$

4 Diffraction in Optical Instruments

4.1 Young's Double Slit (Equal Widths)

For a double slit with equal slit width a and separation d (centre-to-centre), the intensity pattern is

$$I = I_0 \cos^2\left[\frac{kd\sin\theta}{2}\right]\mathrm{sinc}^2\left[\frac{ka\sin\theta}{2}\right]. \tag{72}$$

Two limiting cases:

- $a \to 0$: two coherent point sources, $I = I_0\cos^2\left(\frac{kd\sin\theta}{2}\right)$.
- $d \to 0$: single 1D aperture, $I = I_0\mathrm{sinc}^2\left(\frac{ka\sin\theta}{2}\right)$.

This result can be obtained as a special case ($w_1 = w_2 = a$) of the more general unequal-width double slit discussed in the next subsection.

4.2 Double Slit Interference with Unequal Slit Widths (General Case)

Let the two slits have widths w_1 and w_2, separated by a distance d (centre-to-centre). The field integral is

$$\begin{aligned}
u_P &\propto \int e^{ikr'}\,\mathrm{d}y \\
&\propto e^{ikr}\int e^{-iky\sin\theta}\,\mathrm{d}y \\
&\propto \frac{e^{ikr}}{-ik\sin\theta}\left[\int_{-\frac{d}{2}-\frac{w_2}{2}}^{-\frac{d}{2}+\frac{w_2}{2}}\mathrm{d}y\; e^{-iky\sin\theta} + \int_{\frac{d}{2}-\frac{w_1}{2}}^{\frac{d}{2}+\frac{w_1}{2}}\mathrm{d}y\; e^{-iky\sin\theta}\right] \\
&\propto e^{ikr}\left[e^{i\frac{kd\sin\theta}{2}}(-2i)\frac{\sin\frac{kw_2\sin\theta}{2}}{-ik\sin\theta} + e^{-i\frac{kd\sin\theta}{2}}(-2i)\frac{\sin\frac{kw_1\sin\theta}{2}}{-ik\sin\theta}\right].
\end{aligned} \tag{73}$$

Simplifying gives the complex amplitude

$$u_P \propto e^{ikr}\left[e^{i\frac{\delta}{2}}w_2\mathrm{sinc}\alpha_2 + e^{-i\frac{\delta}{2}}w_1\mathrm{sinc}\alpha_1\right], \tag{74}$$

with

$$\delta = kd\sin\theta, \qquad \alpha_1 = \frac{kw_1\sin\theta}{2}, \qquad \alpha_2 = \frac{kw_2\sin\theta}{2}. \tag{75}$$

The intensity $I_P \propto |u_P|^2$ expands to

$$\begin{aligned}
I_P &\propto \left(e^{i\frac{\delta}{2}}w_2\mathrm{sinc}\alpha_2 + e^{-i\frac{\delta}{2}}w_1\mathrm{sinc}\alpha_1\right)\left(e^{-i\frac{\delta}{2}}w_2\mathrm{sinc}\alpha_2 + e^{i\frac{\delta}{2}}w_1\mathrm{sinc}\alpha_1\right) \\
&\propto w_1^2\mathrm{sinc}^2\alpha_1 + w_2^2\mathrm{sinc}^2\alpha_2 + w_1w_2\mathrm{sinc}\alpha_1\mathrm{sinc}\alpha_2\left(e^{i\delta} + e^{-i\delta}\right) \\
&= w_1^2\mathrm{sinc}^2\alpha_1 + w_2^2\mathrm{sinc}^2\alpha_2 + 2w_1w_2\mathrm{sinc}\alpha_1\mathrm{sinc}\alpha_2\cos\delta.
\end{aligned} \tag{76}$$

Thus the final intensity formula is

$$I_P = I_1\mathrm{sinc}^2\alpha_1 + I_2\mathrm{sinc}^2\alpha_2 + 2\sqrt{I_1I_2}\,\mathrm{sinc}\alpha_1\,\mathrm{sinc}\alpha_2\cos\delta, \tag{77}$$

where

$$\delta = kd\sin\theta, \qquad \alpha_1 = \frac{kw_1\sin\theta}{2}, \qquad \alpha_2 = \frac{kw_2\sin\theta}{2}, \tag{78}$$

and $I_1 \propto w_1^2,\ I_2 \propto w_2^2$. Equivalently, with $\beta = \frac{\delta}{2} = \frac{kd\sin\theta}{2}$ and using $\cos\delta = 2\cos^2\beta - 1$, we have

$$I_P = I_1\mathrm{sinc}^2\alpha_1 + I_2\mathrm{sinc}^2\alpha_2 + 2\sqrt{I_1 I_2}\,\mathrm{sinc}\alpha_1\,\mathrm{sinc}\alpha_2\left(2\cos^2\beta - 1\right), \tag{79}$$

with

$$\beta = \frac{\pi d}{\lambda}\sin\theta, \qquad \alpha_1 = \frac{\pi w_1}{\lambda}\sin\theta, \qquad \alpha_2 = \frac{\pi w_2}{\lambda}\sin\theta. \tag{80}$$

When $w_1 = w_2 = a$, this reduces to the standard Young's double-slit formula given in §4.2.

4.3 Diffraction in Optical Instruments

4.3.1 Simple Imaging Lens

Lens equation:

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}. \tag{81}$$

4.3.2 Telescope

Angular resolution:

$$\Delta\theta = 1.22\frac{\lambda}{D}. \tag{82}$$

Resolving power:

$$R = \frac{1}{\Delta\theta} \approx \frac{D}{1.22\lambda}. \tag{83}$$

4.3.3 Microscope

Define optical path lengths:

$$S_P = nu + S_{\text{lens}} + v, \tag{84}$$

$$S_Q = n\sqrt{u^2 + d^2} + S_{\text{lens}} + \sqrt{v^2 + d'^2}, \tag{85}$$

$$S_{P,M} = n\sqrt{u^2 + \left(\frac{D}{2}\right)^2} + \sqrt{v^2 + \left(\frac{D}{2}\right)^2}, \tag{86}$$

$$S_{Q,M} = n\sqrt{u^2 + \left(\frac{D}{2} + d\right)^2} + \sqrt{v^2 + \left(\frac{D}{2} - d'\right)^2}. \tag{87}$$

The imaging condition requires $S_P = S_{P,M}$ and $S_Q = S_{Q,M}$. In the far-field limit ($u, v \gg D \gg d', d$), we obtain

$$n\frac{Dd}{\sqrt{u^2 + (D/2)^2}} = \frac{Dd'}{\sqrt{v^2 + (D/2)^2}}. \tag{88}$$

Simplified to

$$nd\sin\theta = d'\sin\theta', \tag{89}$$

assuming the image space is air ($n' = 1$). With the geometric approximations

$$\frac{d'}{v} = \tan\alpha \approx \sin\alpha \approx 1.22\frac{\lambda}{D}, \qquad \sin\theta' \approx \frac{D}{2v}, \tag{90}$$

we obtain the resolution criteria.

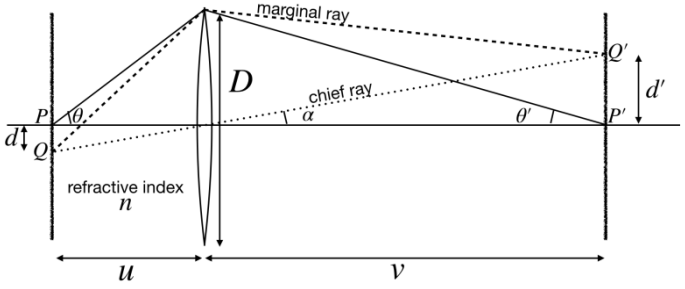


Figure 5: Schematic of a compound microscope: the objective lens forms a magnified real image, which is further magnified by the eyepiece for visual observation.

4.4 Microscope Resolution

Microscope resolution:

$$\text{Linear resolution: } d \approx 0.61\frac{\lambda}{n\sin\theta}, \tag{91}$$

$$\text{Resolving power: } R = \frac{1}{d} = \frac{\text{NA}}{0.61\lambda}, \tag{92}$$

$$\text{Numerical aperture: } \text{NA} = n\sin\theta. \tag{93}$$

4.5 Diffraction Grating

The intensity profile for a grating with N slits, slit width a, and period d is obtained by multiplying the interference factor from §2.5 (point sources) with the single-slit diffraction envelope:

$$I(\theta) = I_0\left[\frac{\sin(N\beta)}{\sin\beta}\right]^2 \text{sinc}^2\alpha, \tag{94}$$

where

$$\beta = \frac{kd\sin\theta}{2} = \frac{\pi d}{\lambda}\sin\theta, \qquad \alpha = \frac{ka\sin\theta}{2} = \frac{\pi a}{\lambda}\sin\theta. \tag{95}$$

For a transmission grating with $a = d/2$, the maxima condition is

$$\frac{kd\sin\theta}{2} = n\pi \quad \Longrightarrow \quad n\lambda = d\sin\theta. \tag{96}$$

Angular dispersion:

$$\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} = \frac{n}{d\cos\theta}. \tag{97}$$

Free spectral range:

$$\Delta\lambda_{\text{FSR}} = \lambda_{\text{red}} - \lambda_{\text{blue}}, \qquad n\lambda_{\text{red}} = (n+1)\lambda_{\text{blue}} \quad \Longrightarrow \quad \Delta\lambda_{\text{FSR}} = \frac{\lambda_{\text{blue}}}{n}. \tag{98}$$

Rayleigh's criterion: the diffraction maximum of one wavelength coincides with the first minimum of the adjacent wavelength.

$$\text{Maxima: } \frac{kd\sin\theta}{2} = n\pi \quad \Longrightarrow \quad d\sin\theta = n\lambda, \tag{99}$$

$$\text{Minima: } \frac{Nkd\sin\theta'}{2} = m\pi \quad \Longrightarrow \quad d\sin\theta' = \frac{m\lambda}{N}. \tag{100}$$

Set $m = nN + 1$. Then

$$\Delta\sin\theta = \sin\theta' - \sin\theta = \frac{\lambda}{d}\left(\frac{m}{N} - n\right) = \frac{\lambda}{Nd}, \tag{101}$$

$$\Delta\lambda = \Delta\left(\frac{d\sin\theta}{n}\right) = \frac{d}{n}\cos\theta\,\Delta\theta = \frac{d}{n}\Delta\sin\theta. \tag{102}$$

Thus the resolving power is

$$R = \frac{\lambda}{\Delta\lambda} = \frac{d\sin\theta}{n}\cdot\frac{1}{\frac{d}{Nd}\Delta\sin\theta} \quad \Longrightarrow \quad R = Nn. \tag{103}$$

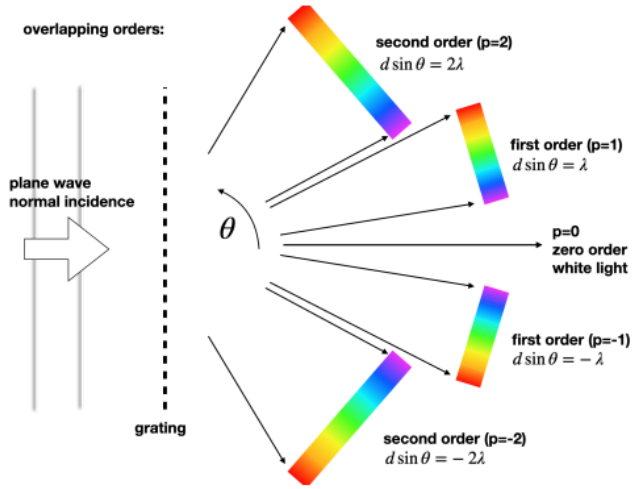


Figure 6: Diffraction grating principle: a periodic array of slits produces sharp interference maxima; the angular position of each order is determined by the grating equation $d\sin\theta = n\lambda$.

5 Interferometry

5.1 Interference Fringes

5.1.1 Equal inclination fringes

The optical path difference is

$$\Delta L = 2n_2 d\cos\theta_r. \tag{104}$$

With phase shift $\delta = k_0\Delta L - \pi$, where $k_0 = \frac{2\pi}{\lambda_2}$, the intensity is

$$I = I_0\cos^2\left(\frac{\delta}{2}\right). \tag{105}$$

Maxima occur when $\frac{\delta}{2} = m\pi$, i.e.,

$$d\cos\theta_r = \frac{(2m+1)\lambda_2}{4}. \tag{106}$$

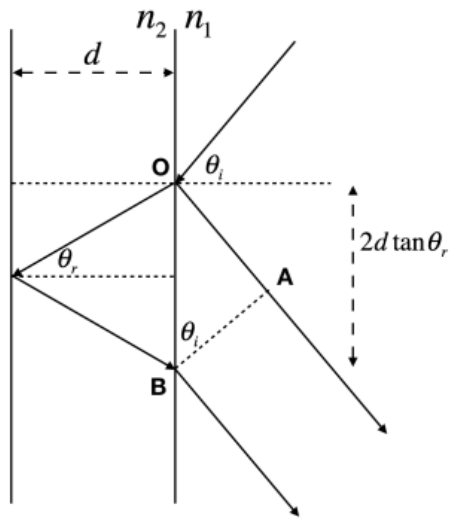


Figure 7: Equal-inclination fringes (Haidinger fringes): circular fringes observed when a parallel plate is illuminated by an extended monochromatic source.

5.1.2 Equal thickness fringes

For normal incidence, $\theta_i = 0 \Rightarrow \theta_r = 0$. If $d = \alpha x$, then

$$d\cos\theta_r = \frac{(2m+1)\lambda_2}{4} \quad \Longrightarrow \quad x = \frac{(2m+1)\lambda_2}{4\alpha}. \tag{107}$$

5.2 The Michelson Interferometer

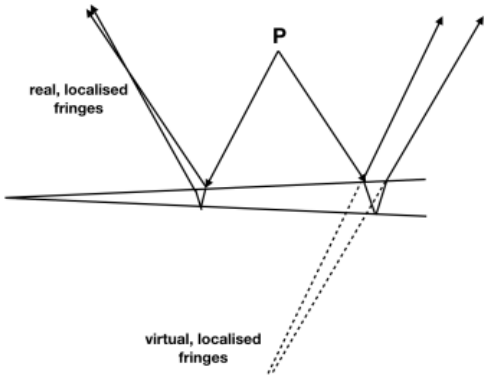


Figure 8: Equal-thickness fringes (Fizeau fringes): straight or curved fringes corresponding to contours of constant film thickness in a wedge-shaped thin film.

Fringe geometry The optical path difference is $\Delta L = 2x\cos\theta$, and the phase difference is

$$\delta = k\Delta L = 2kx\cos\theta. \tag{108}$$

Maxima conditions:

$$2x = m_0\lambda, \qquad 2x\cos\theta_p = (m_0 - p)\lambda \quad (p\text{-th ring}). \tag{109}$$

For small angles $\theta_p \ll \pi/2$, $\cos\theta_p \approx 1 - \theta_p^2/2$, we get

$$2x(1-\cos\theta_p) = p\lambda \quad \Longrightarrow \quad \theta_p^2 = p\,\frac{\lambda}{x}. \tag{110}$$

With $\theta = r/f$, the ring spacing is

$$\theta_{p+1}^2 - \theta_p^2 = \frac{\lambda}{x}, \qquad r_{p+1}^2 - r_p^2 = \frac{f^2\lambda}{x}. \tag{111}$$

Spectral resolution On axis ($\theta = 0$):

$$u(x) \propto \left(1 + e^{i2kx}\right) \quad \Longrightarrow \quad I(x) \propto \left(1 + \cos(2kx)\right). \tag{112}$$

Thus

$$I(x) = \frac{1}{2} I_{\max}\left(1 + \cos(2kx)\right). \tag{113}$$

For two spectral lines λ_1, λ_2 with k_1, k_2:

$$\begin{aligned} I(x) &= \frac{1}{2} I_{\max}\left[1 + \cos(2k_1x) + 1 + \cos(2k_2x)\right] \\ &= I_{\max}\left\{1 + \cos\left[(k_1+k_2)x\right]\cos\left[(k_1-k_2)x\right]\right\}. \end{aligned} \tag{114}$$

The minimum resolvable wavevector difference is

$$\Delta k = k_1 - k_2 = \frac{\pi}{x_{\max}} = \frac{2\pi}{L_{\max}}, \tag{115}$$

where $L_{\max} = 2x_{\max}$ is the maximum optical path difference. The resolving power is

$$R = \frac{\lambda}{\Delta\lambda} = \frac{k}{\Delta k} = \frac{L_{\max}}{\lambda}. \tag{116}$$

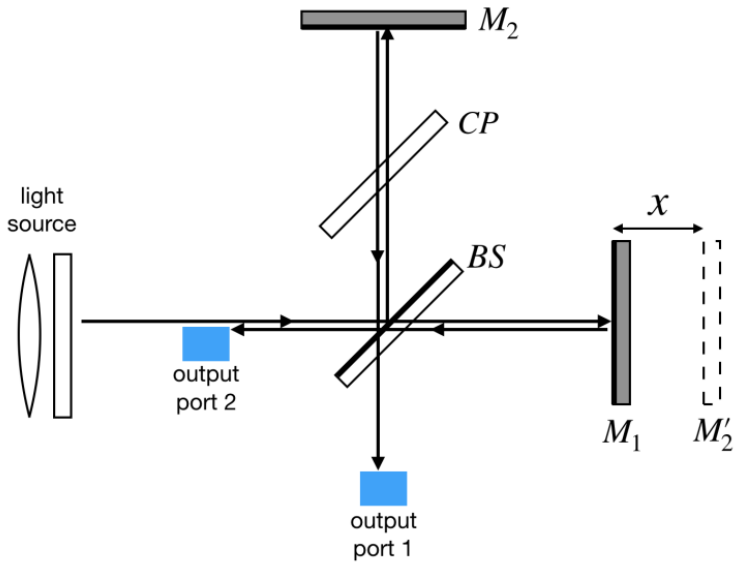


Figure 9: Michelson interferometer: a beam splitter divides the incident light into two paths; the recombined beams produce interference fringes whose visibility depends on the optical path difference.

5.3 The Fabry–Perot Etalon and Spectrometer

In contrast to the two-beam interference of the Michelson interferometer, the Fabry–Perot etalon exploits multiple-beam interference between two parallel partially reflecting surfaces. This yields much sharper transmission peaks and consequently a significantly higher resolving power.

Let the reflection and transmission coefficients at the interfaces be r, t (from n_1 to n_2) and r', t' (from n_2 to n_1). The reflected and transmitted intensities are $I_R = |u_R|^2$, $I_T = |u_T|^2$. The optical path difference is

$$\Delta L = 2n_2 d\cos\theta_t, \tag{117}$$

and phase difference $\delta = k\Delta L = 2kn_2 d\cos\theta_t$. Reciprocity relations: $r = -r'$, $r^2 + tt' = 1$; reflectance and transmittance are $R = r^2$, $T = tt'$.

The reflected field is

$$u_R = ur + utt'r'e^{i\delta}\left(1 + r'^2 e^{i\delta} + r'^4 e^{i2\delta} + \cdots\right) = ur\,\frac{1 - e^{i\delta}}{1 - r^2 e^{i\delta}}. \tag{118}$$

The transmitted field is

$$u_T = utt' e^{ikd/\cos\theta_t}\left(1 + r^2 e^{i\delta} + r^4 e^{i2\delta} + \cdots\right) = ue^{ikd/\cos\theta_t}\,\frac{T}{1 - Re^{i\delta}}. \tag{119}$$

The intensity formulas:

$$I_R = I \cdot \frac{4R\sin^2(\delta/2)}{(1-R)^2 + 4R\sin^2(\delta/2)}, \tag{120}$$

$$I_T = I \cdot \frac{1}{1 + \mathcal{F}\sin^2(\delta/2)}, \tag{121}$$

where the finesse coefficient is

$$\mathcal{F} = \frac{4R}{(1-R)^2}. \tag{122}$$

Transmission maxima occur when $\delta = 2\pi m$, i.e.,

$$2kn_2 d\cos\theta_t = 2\pi m \quad \Longrightarrow \quad \cos\theta_t = \frac{m\lambda}{2n_2 d}. \tag{123}$$

For small angles $0 < \theta_t \ll \pi/2$, the angular separation of fringes is

$$\theta_m^2 - \theta_{m+1}^2 = \frac{\lambda}{n_2 d}. \tag{124}$$

The finesse and resolving power are

$$F = \frac{\pi\sqrt{R}}{1-R}, \qquad R = \frac{\lambda}{\Delta\lambda} = mF. \tag{125}$$

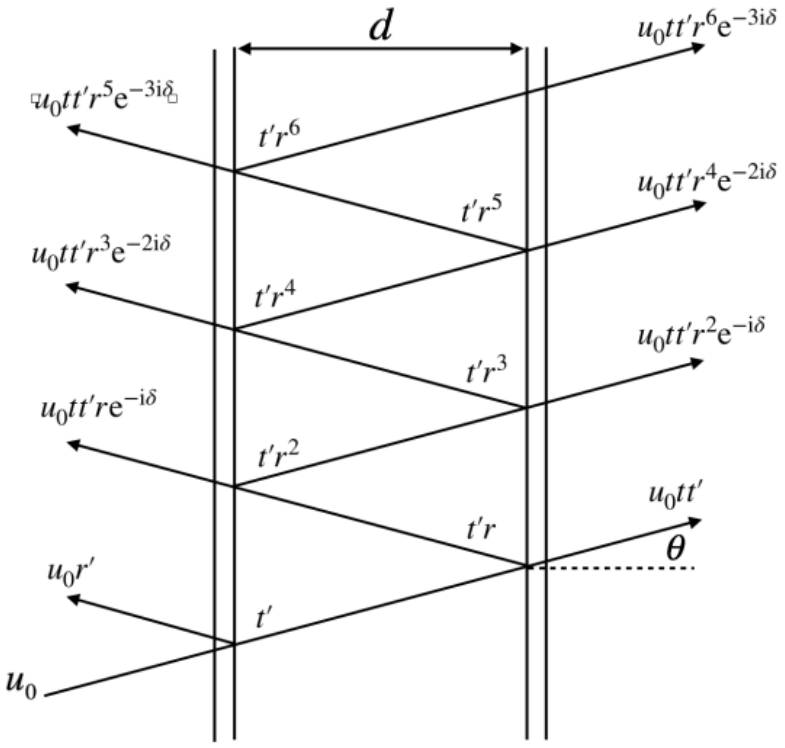


Figure 10: Fabry–Perot etalon: multiple-beam interference between two parallel partially reflective surfaces produces sharp transmission peaks, with resolving power proportional to the finesse and the order m.

6 Fundamentals of Lasers

6.1 Interaction of Atoms

The radiative transition rates per unit volume are

$$\gamma_{21} = A_{21}n_2 + B_{21}u(\omega)n_2 \quad \text{(spontaneous + stimulated emission)}, \tag{126}$$
$$\gamma_{12} = B_{12}u(\omega)n_1 \quad \text{(stimulated absorption)}. \tag{127}$$

The photon energy is $\hbar\omega = E_2 - E_1$. In thermal equilibrium, the population ratio is

$$\frac{n_2}{n_1} = \frac{g_2}{g_1}\exp\left(-\frac{\hbar\omega}{k_BT}\right). \tag{128}$$

The blackbody spectral energy density is

$$u(\omega) = \frac{\hbar\omega^3}{\pi^2c^3}\frac{1}{e^{\hbar\omega/k_BT}-1}. \tag{129}$$

Detailed balance in equilibrium ($\gamma_{21} = \gamma_{12}$, $A_{21} = A$) gives

$$A = \left(B_{12}\frac{n_1}{n_2} - B_{21}\right)u(\omega). \tag{130}$$

Substituting the Boltzmann distribution:

$$A = \left[B_{12}\frac{g_1}{g_2}\exp\left(\frac{\hbar\omega}{k_BT}\right) - B_{21}\right]\frac{\hbar\omega^3}{\pi^2c^3}\frac{1}{e^{\hbar\omega/k_BT}-1}. \tag{131}$$

Since A, B_{12}, B_{21} are temperature-independent, we require $B_{12}g_1 = B_{21}g_2$. Hence the Einstein relations:

$$A_{21} = B_{12}\frac{g_1}{g_2}\frac{\hbar\omega^3}{\pi^2c^3} = B_{21}\frac{\hbar\omega^3}{\pi^2c^3}. \tag{132}$$

Final form:

$$\frac{A_{21}}{B_{21}} = \frac{\hbar\omega^3}{\pi^2c^3}, \qquad B_{12}g_1 = B_{21}g_2. \tag{133}$$

6.2 Conditions for Laser Gain

Net stimulated emission requires

$$n_2B_{21}u(\omega) > n_1B_{12}u(\omega) \quad \Longrightarrow \quad n_2 > n_1. \tag{134}$$

If $g_1 = g_2$, then $B_{21} = B_{12}$. The gain threshold is $B_{21}u(\omega) > A_{21}$.

Pumping Let τ_1, τ_2 be the excited-state lifetimes, and R_1, R_2 the pump rates into the lower and upper levels. The rate equations are

$$\frac{\mathrm{d}n_2}{\mathrm{d}t} = R_2 - \frac{n_2}{\tau_2}, \tag{135}$$
$$\frac{\mathrm{d}n_1}{\mathrm{d}t} = R_1 + n_2A_{21} - \frac{n_1}{\tau_1}. \tag{136}$$

In the steady state ($\mathrm{d}n_1/\mathrm{d}t = \mathrm{d}n_2/\mathrm{d}t = 0$):

$$n_2 = R_2\tau_2, \tag{137}$$

$$n_1 = R_1\tau_1 + R_2\tau_2 A_{21}\tau_1. \tag{138}$$

The population inversion condition $n_2 > n_1$ yields

$$\frac{R_2\tau_2}{R_1\tau_1}\left(1 - A_{21}\tau_1\right) > 1. \tag{139}$$

Thus two inequalities are required:

$$R_2\tau_2 > R_1\tau_1, \qquad A_{21}\tau_1 < 1. \tag{140}$$

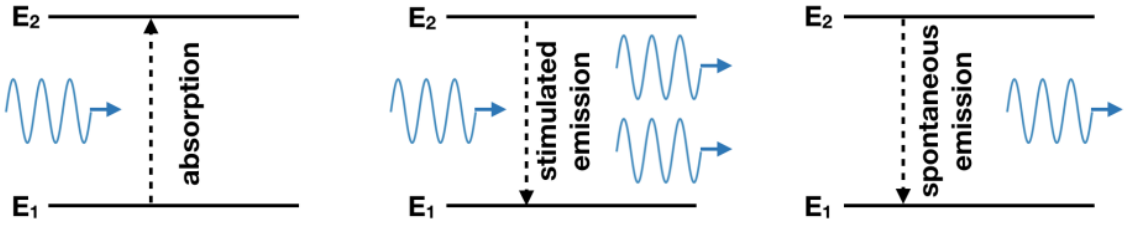


Figure 11: Laser operation principle: population inversion between two energy levels enables stimulated emission to dominate over absorption, producing coherent optical amplification.

6.3 Laser Longitudinal Modes

6.3.1 Resonance Condition

$$2L = m\lambda, \tag{141}$$

where L is the cavity length and m is the mode number.

6.3.2 Frequency Separation of Consecutive Longitudinal Modes

$$\nu = \frac{c}{\lambda} = \frac{c}{2L}m. \tag{142}$$

The separation is

$$\Delta\nu = \nu(m+1) - \nu(m) = \frac{c}{2L}. \tag{143}$$

6.3.3 Wavelength Linewidth

$$\Delta\nu_{\text{line}} = \frac{c}{\lambda^2}\Delta\lambda_{\text{line}}, \tag{144}$$

where λ is the laser wavelength and $\Delta\lambda_{\text{line}}$ is the spontaneous emission linewidth.

6.3.4 Number of Lasing Longitudinal Modes

$$N \approx \frac{\Delta\nu_{\text{line}}}{\Delta\nu}. \tag{145}$$

7 Coherence Theory

7.1 Coherence

Classification:

- Temporal coherence: phase correlation over time.
- Spatial (transverse) coherence: phase correlation at different spatial positions.

7.1.1 Temporal Coherence

The spectral bandwidth limits temporal coherence. The coherence time and length are

$$\tau_c = \frac{1}{\Delta\nu}, \qquad L_c = c\tau_c = \frac{\lambda^2}{\Delta\lambda}. \tag{146}$$

In a Michelson interferometer, the fringe visibility is

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}. \tag{147}$$

Limits:

- $V = 1$: perfect coherence,
- $V = 0$: complete incoherence,
- $0 < V < 1$: partial coherence.

With the complex degree of coherence $\gamma = \frac{\langle u_A u_B^* \rangle}{\sqrt{I_A I_B}}$, the intensity is

$$I = I_A + I_B + 2\sqrt{I_A I_B}\ \mathrm{Re}[\gamma]. \tag{148}$$

The visibility is related to γ by

$$V = \frac{2\sqrt{I_A I_B}}{I_A + I_B}\,|\gamma|. \tag{149}$$

7.1.2 Spatial (Transverse) Coherence

1) Intensity from two symmetric coherent sources:

$$I(\theta) = I_0\left[1 + \cos(ka\sin\theta)\cos(ka\sin\theta_s)\right]. \tag{150}$$

Alternative forms:

$$I(\theta) = I_0\cos^2\left(\frac{ka(\sin\theta \mp \sin\theta_s)}{2}\right) = \frac{I_0}{2}\left(1 + \cos\left[ka(\sin\theta \mp \sin\theta_s)\right]\right). \tag{151}$$

The total intensity is $I(\theta) = I(\theta)_+ + I(\theta)_-$.

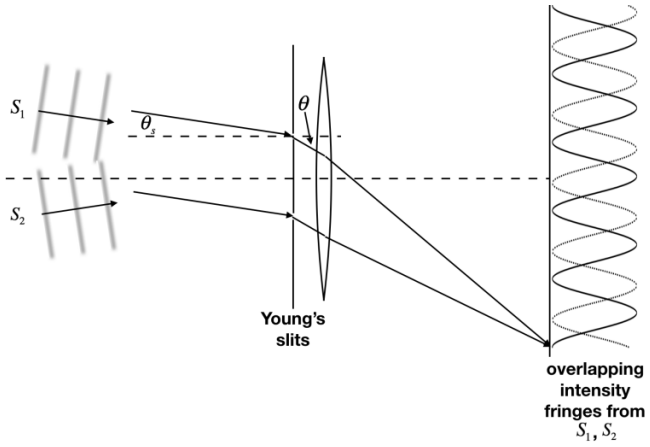


Figure 12: Transverse coherence: the degree of coherence between two points in the observation plane is determined by the angular size of the source, as described by the Van Cittert–Zernike theorem.

7.2 Extended Source Spatial Coherence

For two point sources, the maximum and minimum intensities are

$$I_{\max} = I_0 \left(1 + \cos(ka \sin \theta_s)\right), \qquad I_{\min} = I_0 \left(1 - \cos(ka \sin \theta_s)\right). \tag{152}$$

Thus

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \cos(ka \sin \theta_s). \tag{153}$$

In the point-source limit $\theta_s \to 0$, we get $V \to 1$, $I_{\max} \to 2I_0$, $I_{\min} \to 0$, sharp fringes.

2) Extended uniform source $-\Theta_s < \theta_s < \Theta_s$. Let $k' = k \sin \theta_s$. The intensity is proportional to

$$I(\theta) \propto \int_0^{k \sin \Theta_s} \left[1 + \cos(ka \sin \theta) \cos(k'a)\right] \mathrm{d}k', \tag{154}$$

where we have used symmetry to integrate only half the range. Evaluation gives

$$I(\theta) \propto k \sin \Theta_s + \cos(ka \sin \theta) \cdot \frac{\sin\left(ka \sin \Theta_s\right)}{a}. \tag{155}$$

Compact form:

$$I(\theta) = I_0 \left[1 + \cos(ka \sin \theta) \cdot \mathrm{sinc}\left(ka \sin \Theta_s\right)\right]. \tag{156}$$

The extrema are

$$I_{\max} = I_0 \left[1 + \mathrm{sinc}(ka \sin \Theta_s)\right], \qquad I_{\min} = I_0 \left[1 - \mathrm{sinc}(ka \sin \Theta_s)\right]. \tag{157}$$

Hence the visibility is

$$V = \mathrm{sinc}\left(ka \sin \Theta_s\right). \tag{158}$$

7.3 Van Cittert–Zernike Theorem

The theorem states that the complex degree of coherence between two points in the interference plane equals the Fourier transform of the source intensity distribution.

7.3.1 Astronomical Imaging Interference

For a single telescope of diameter D:

$$\theta_{\min} = 1.22\frac{\lambda}{D}. \tag{159}$$

For an interferometer with baseline B:

$$\theta_{\min} \approx \frac{\lambda}{B}. \tag{160}$$

8 Polarization of Light

8.1 General Formulation

The orthogonal field components are

$$E_x = E_{x0}\cos(\omega t - kz + \phi_x), \qquad E_y = E_{y0}\cos(\omega t - kz + \phi_y). \tag{161}$$

Define $\delta = \phi_y - \phi_x$. The polarization ellipse equation is

$$\left(\frac{E_x}{E_{x0}}\right)^2 + \left(\frac{E_y}{E_{y0}}\right)^2 - 2\frac{E_x E_y}{E_{x0}E_{y0}}\cos\delta = \sin^2\delta. \tag{162}$$

8.2 Polarization States

1. Linear polarization: $\delta = 0$ or $\delta = \pi$.
2. Circular polarization: $\delta = \pm\frac{\pi}{2}$, $E_{x0} = E_{y0}$.
   - $\delta > 0$: right-handed circular,
   - $\delta < 0$: left-handed circular.
3. Elliptical polarization: all other phase/amplitude combinations.

8.3 Jones Vectors

The general Jones vector is

$$\mathbf{E} = \begin{pmatrix} E_{x0} \\ E_{y0} \end{pmatrix} = E_x \begin{pmatrix} 1 \\ |E_y/E_{x0}|e^{i\delta} \end{pmatrix}. \tag{163}$$

Standard polarization vectors:

$$\text{Linear } x\text{-axis: } \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \tag{164}$$

$$\text{Linear } y\text{-axis: } \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \tag{165}$$

$$\text{Linear } 45^\circ: \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \tag{166}$$

$$\text{Right circular: } \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}, \tag{167}$$

$$\text{Left circular: } \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}. \tag{168}$$

8.4 Jones Matrices

The field transformation is $\mathbf{E}_{\text{out}} = J\,\mathbf{E}_{\text{in}}$. Standard matrices:

$$J_{\text{LP},x} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad J_{\text{LP},y} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad J_{\text{LP},45^\circ} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \tag{169}$$

$$J_{\text{HWP}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad J_{\text{QWP}} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad J_{\text{WP}} = \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\delta} \end{pmatrix}. \tag{170}$$

The rotation matrix (counterclockwise rotation by θ) is

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}. \tag{171}$$

8.5 Stokes Parameters

$$I = E_{x0}^2 + E_{y0}^2 \quad \text{(total intensity)}, \tag{172}$$
$$Q = E_{x0}^2 - E_{y0}^2 \quad \text{(horizontal/vertical linear)}, \tag{173}$$
$$U = 2E_{x0}E_{y0}\cos\delta \quad (\pm 45^\circ \text{ linear}), \tag{174}$$
$$V = 2E_{x0}E_{y0}\sin\delta \quad \text{(circular)}. \tag{175}$$

The degree of polarization is

$$P = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{\sqrt{Q^2 + U^2 + V^2}}{I}. \tag{176}$$

8.6 Malus's Law, Brewster Angle, Faraday Effect

8.6.1 Malus's Law

$$I_{\text{out}} = I_{\text{in}} \cos^2\theta, \tag{177}$$

where θ is the angle between the incident linear polarization axis and the polarizer transmission axis.

8.6.2 Brewster's Angle

The Brewster angle θ_B produces purely s-polarized reflected light (perpendicular to the plane of incidence). It satisfies

$$\tan\theta_B = \frac{n_2}{n_1} \iff r_p = \frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)} = 0 \iff \theta_i + \theta_t = \frac{\pi}{2}. \tag{178}$$

8.6.3 Faraday Effect

The rotation angle of linear polarized light is

$$\beta = VBd, \tag{179}$$

where V is the Verdet constant, B the magnetic field strength parallel to propagation, and d the propagation distance.

8.7 Polarization Optical Components and Birefringence

8.7.1 Polarization Optical Components

Linear polarizer (LP):

- Polarized input: $I_{\text{out}} = I_{\text{in}} \cos^2\theta$,
- Unpolarized input: $I_{\text{out}} = I_{\text{in}}/2$.

Wave plates (phase retarders):

1. Half-wave plate (HWP): $\delta = \pi$, $d|n_o - n_e| = \frac{\lambda}{2}$,
2. Quarter-wave plate (QWP): $\delta = \frac{\pi}{2}$, $d|n_o - n_e| = \frac{\lambda}{4}$.

8.7.2 Birefringence

- Ordinary ray (o-ray): obeys Snell's law; $\mathbf{E} \perp$ incident plane (s-polarization); $n_o = \frac{c}{v_o} = \frac{\sin\theta_i}{\sin\theta_o}$ (constant).
- Extraordinary ray (e-ray): does not obey Snell's law; $\mathbf{E} \parallel$ incident plane (p-polarization); $n_e = \frac{c}{v_e} = \frac{\sin\theta_i}{\sin\theta_e}$ (angle-dependent).